LLMOPs ⇒

How we manage, deploy, monitor, and improve AI models like chatgpt in real-world applications.

ex ⇒ If building AI model is building a car, then LLMops is everything needed to run that car safely, such as (Fuel (data), Maintenance (Fine-tuning), Monitoring (performance), upgrades)

What LLMops include?

- Prompt Engineering
- Model monitoring
- Cost optimization
- Version control
- Evaluation
- Deployment

Deeplearning.AI

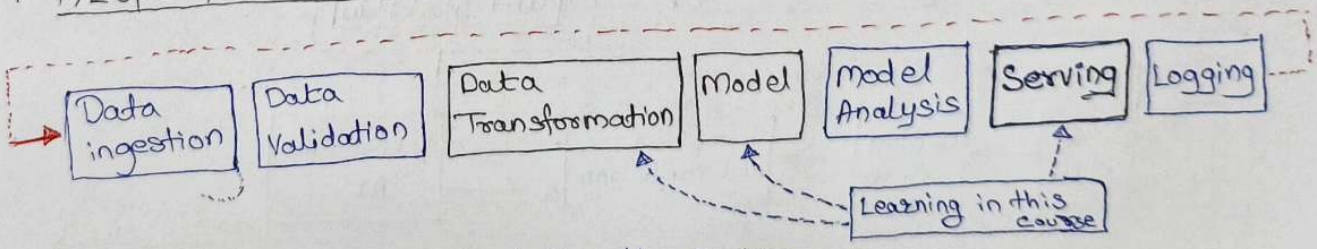
★ What is MLops?

"MLOps is an ML engineering culture & practice that aims at unifying ML development (Dev) and ML operations (Ops)"

Automation and Monitoring at all steps of ML system construction, including:

- Integration
- Testing
- Releasing
- Deployment
- Infrastructure management

→ MLOps framework



Job orchestration :→ we telling the system which step needs to be executed first, then which one after, and so on.

→ MLOps vs System Design for LLM:

MLops for LLMs (LLMOps) :→ Focus on the LLM development and managing the model in production.

- Experiment on foundation models
- Prompt design and management
- Supervised tuning
- Monitoring
- Evaluate

Question it answers :→ How do we ~~ship the~~ build, version, and ship the model reliably?

Contents :→ Data prep and versioning, tuning pipelines, evaluation, deployment, monitoring

Owner :→ ML/platform engineers

LLM System Design :→ Broader design of the entire end-to-end application (front-end, back-end, data engineering, etc)

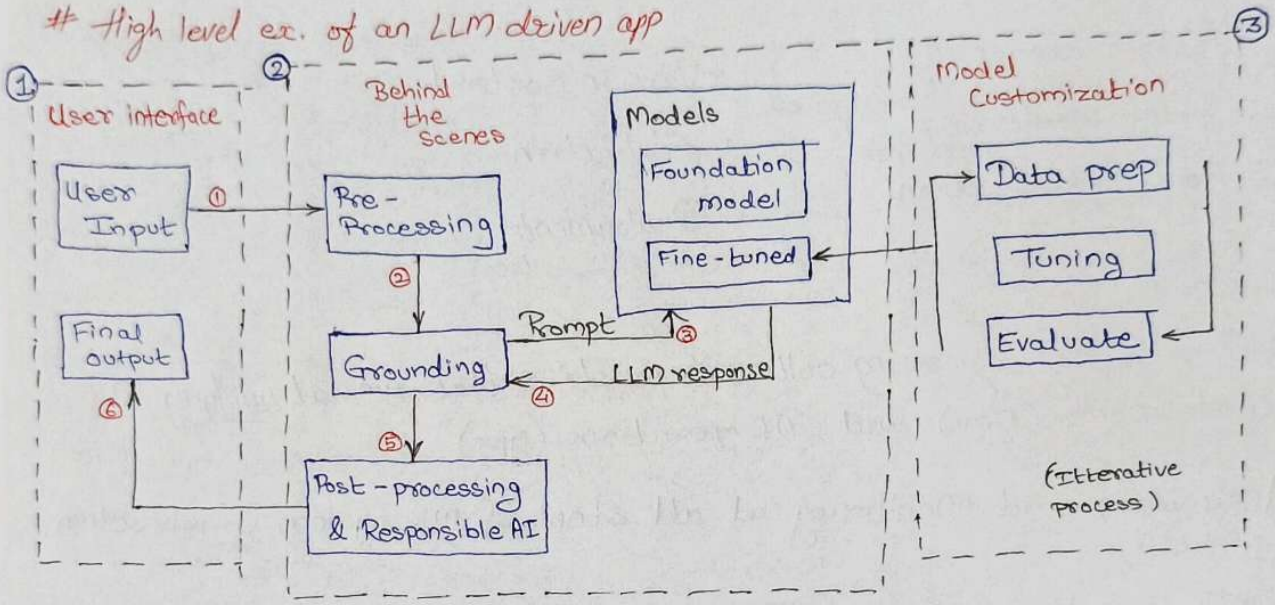
- Chain multiple LLM together
- Grounding
- Track history

Question it :→ How does the app uses the model answers

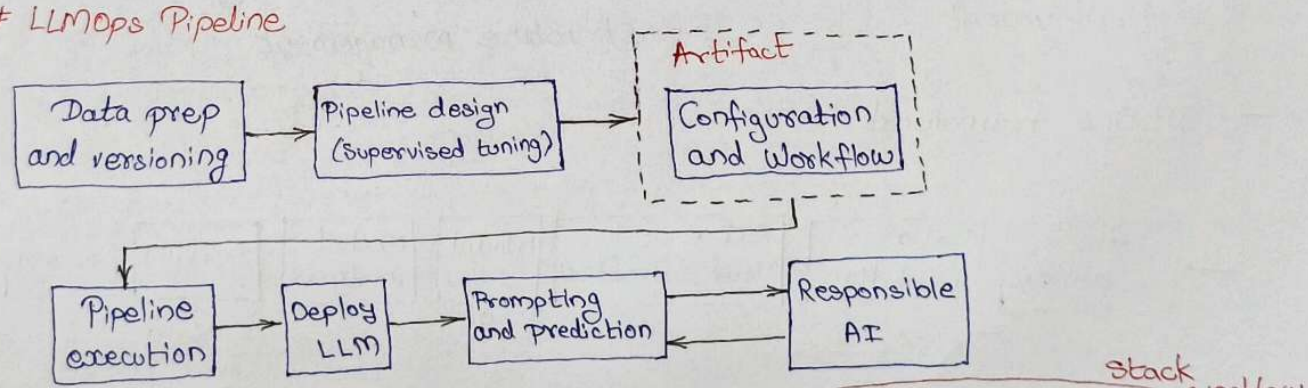
Contents :→ Chaining calls, grounding with external data, keeping chat history, pre- and post-processing

Owner :→ App Engineers

High level ex. of an LLM driven app

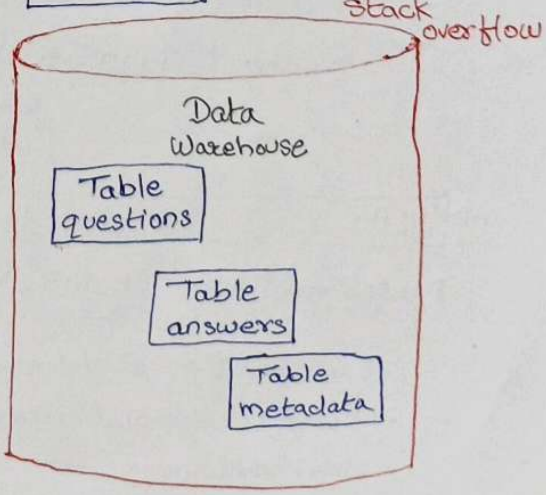


LLMops Pipeline



Data Preparation →

- Generally we store huge amount of data in datawarehouse.
- When needed -
  - we don't load it on our system (bcos of huge data) and perform query.
  - Instead we send the query to the datawarehouse to apply on stored data and usually download the result for further process.



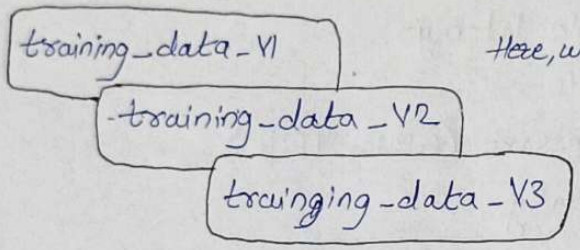
⇒ File formats train and evaluation

JSONL → Ideal choice for small to medium-sized datasets.

TF Record → Binary format and easier to read for computers, making it ideal for efficient training

Parquet → Parquet is a good choice for large and complex datasets.

⇒ Versioning artifacts

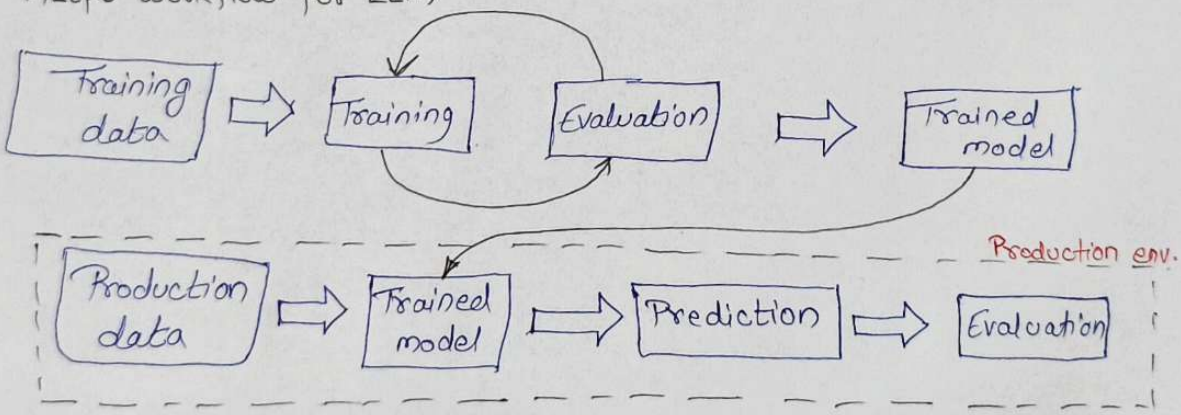


Here, we need to keep track of the data.

It helps to identify from where the LLM response came from.

Automation and Orchestration with Pipelines

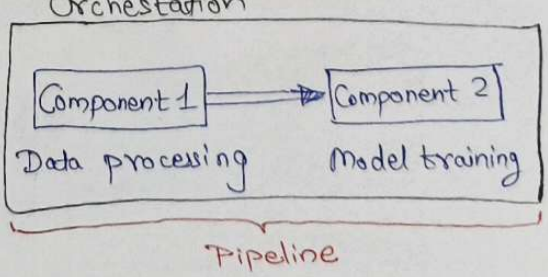
⇒ MLOps workflow for LLM



Orchestation, Automation and Deployment.

Apache airflow

Kubeflow
- DSL: "Domain Specific Language"
- Set of instructions (Vocab/Syntax) for designing pipelines.
- Define the "Configuration" (Pipeline steps



- Rest API

⇒ Prediction, prompts, safety

— Consideration for production deployment

- Package, Deploy and version
- Model monitoring: metrics and safety
- Inference scalability:
  - Load test
  - Controlled roll out, etc.
- Latency: permissible latency
  - Smaller models
  - Faster processors (GPUs, TPUs)
  - Deploy regionally